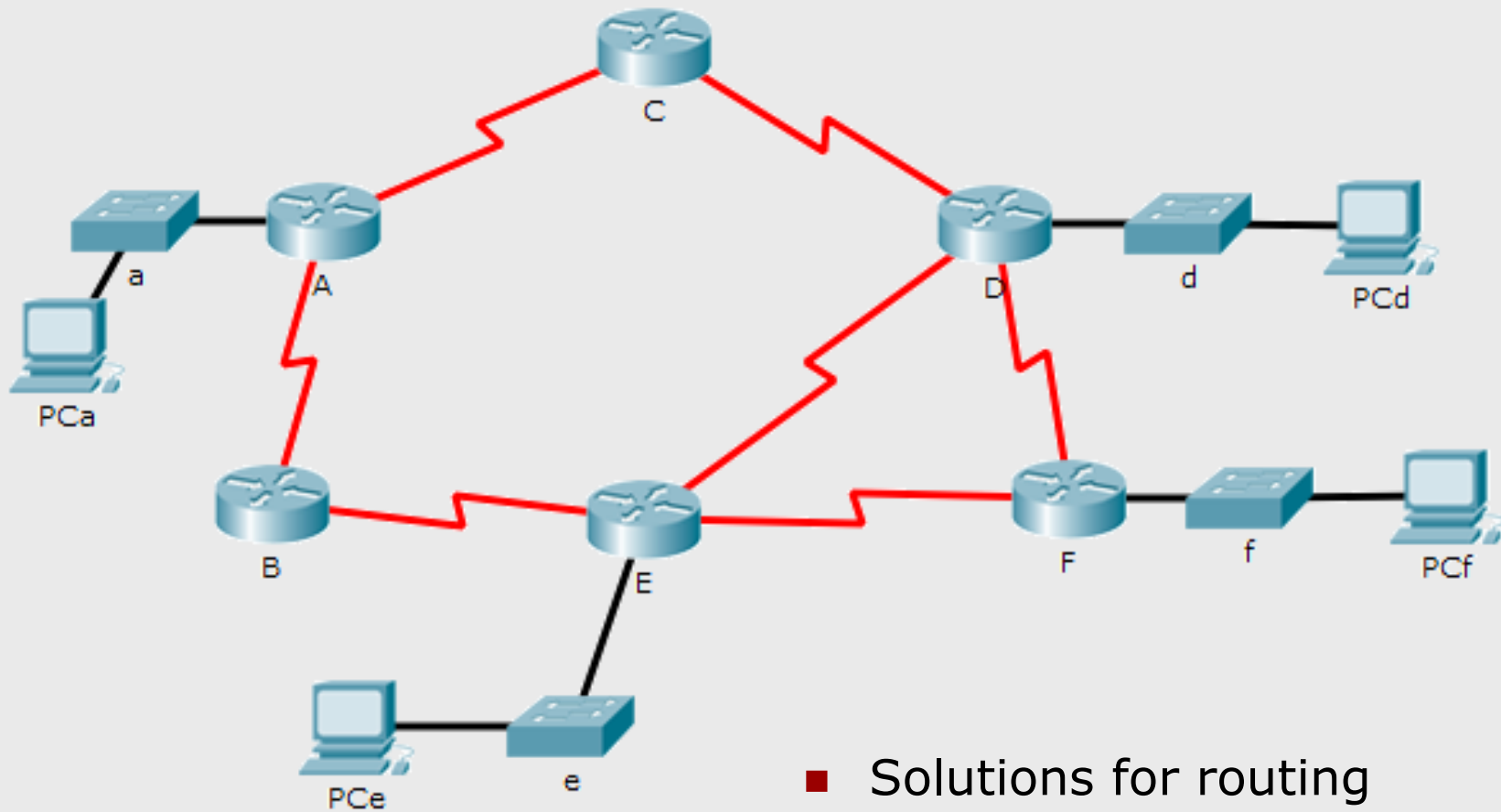


Lab 3

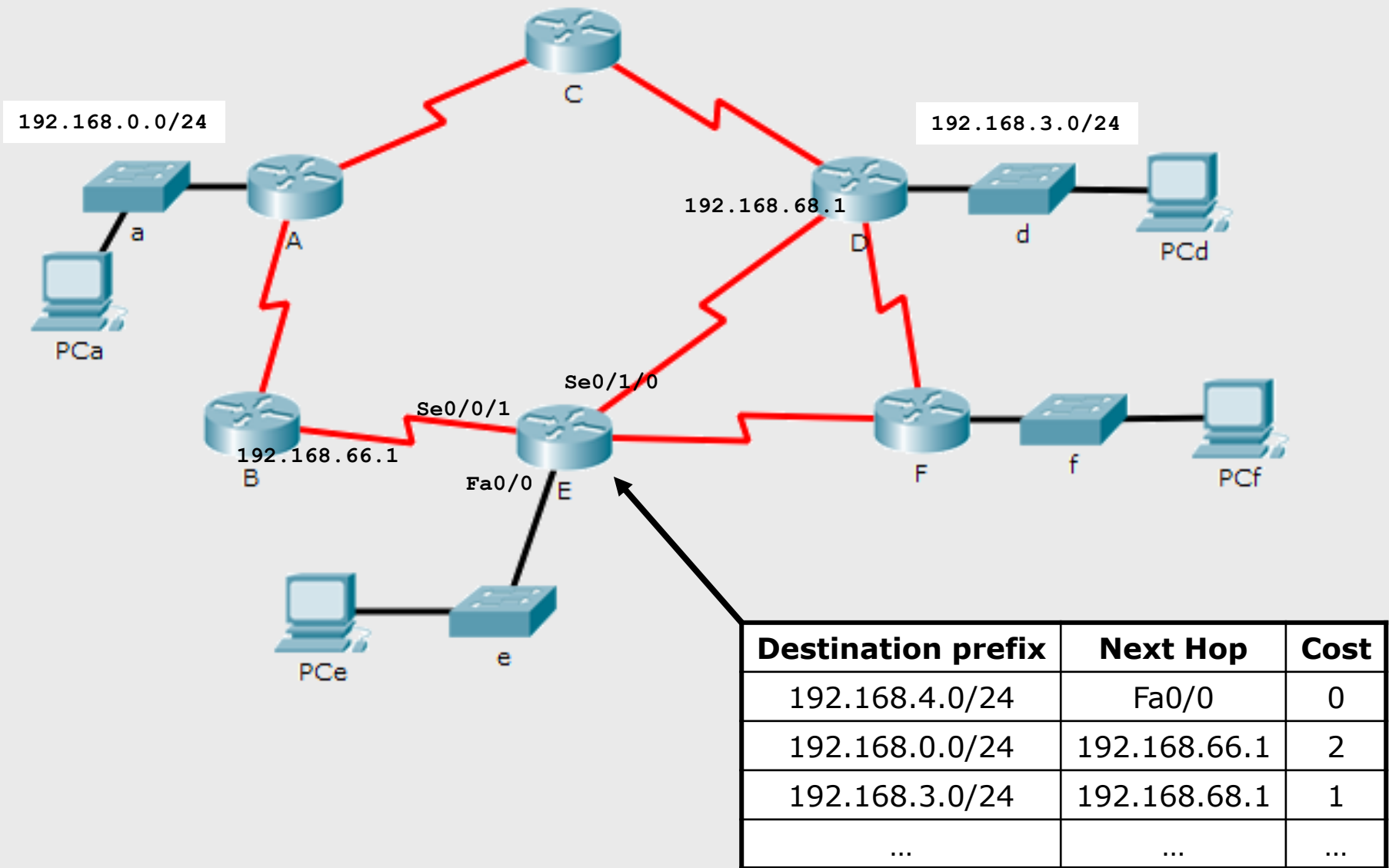
1. IP routing table
2. Static route configuration

Routing of packets



- Solutions for routing
 - *Routing by destination address*
 - *Label swapping*
 - *Source routing*

Routing of packets – IP protocol



Routing table in Cisco IOS

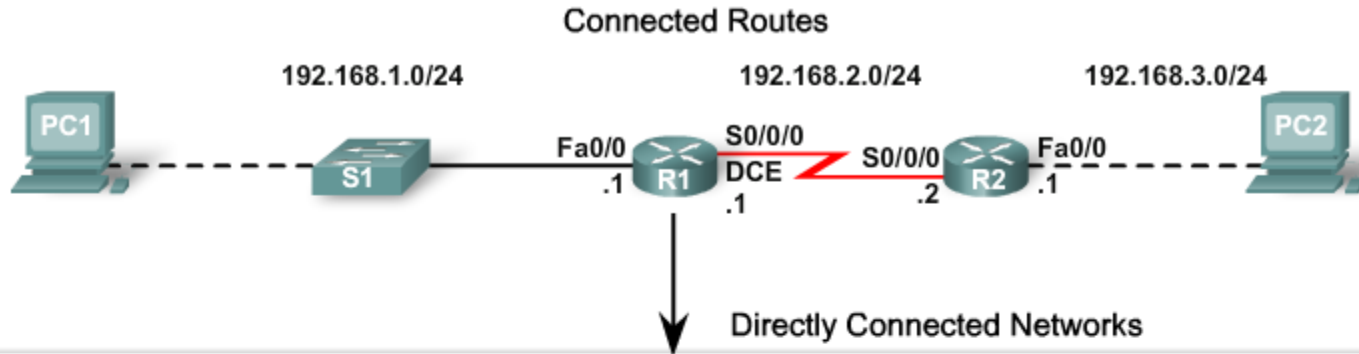
- Stored in RAM, displayed with the **show ip route** command
- Contains information about
 - Directly Connected Routes
 - Static Routes
 - Dynamic Routes

```
RE#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

R 192.168.0.0/24 [120/2] via 192.168.66.1, 00:00:03, Serial0/0/1
R 192.168.3.0/24 [120/1] via 192.168.68.1, 00:00:02, Serial0/1/0
C 192.168.4.0/24 is directly connected, FastEthernet0/0
R 192.168.5.0/24 [120/1] via 192.168.70.2, 00:00:03, Serial0/0/0
R 192.168.64.0/24 [120/1] via 192.168.66.1, 00:00:03, Serial0/0/1
R 192.168.65.0/24 [120/2] via 192.168.68.1, 00:00:02, Serial0/1/0
    [120/2] via 192.168.66.1, 00:00:03, Serial0/0/1
C 192.168.66.0/24 is directly connected, Serial0/0/1
R 192.168.67.0/24 [120/1] via 192.168.68.1, 00:00:02, Serial0/1/0
C 192.168.68.0/24 is directly connected, Serial0/1/0
R 192.168.69.0/24 [120/1] via 192.168.70.2, 00:00:03, Serial0/0/0
    [120/1] via 192.168.68.1, 00:00:02, Serial0/1/0
C 192.168.70.0/24 is directly connected, Serial0/0/0
```

Directly-connected networks



```
R1#show ip route
```

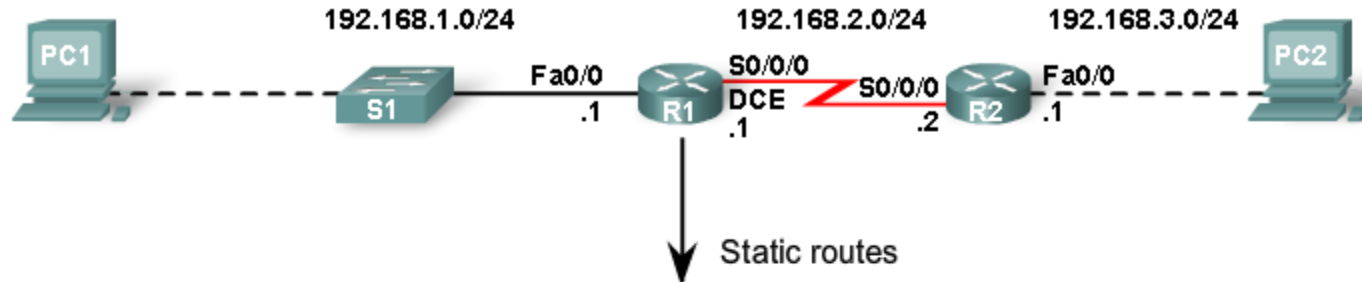
```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0  
C    192.168.2.0/24 is directly connected, Serial0/0/0
```

Static routes

Connected and Static Routes



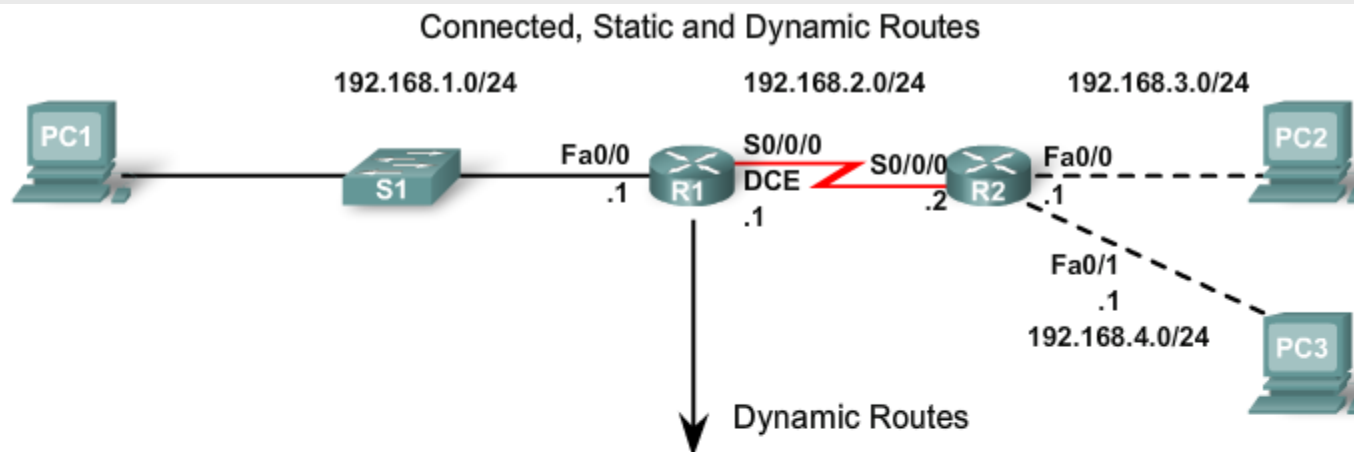
```
R1#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0/0
S    192.168.3.0/24 [1/0] via 192.168.2.2
```

Dynamic routes



```
R1#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
        area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
C    192.168.1.0/24 is directly connected, FastEthernet0/0
```

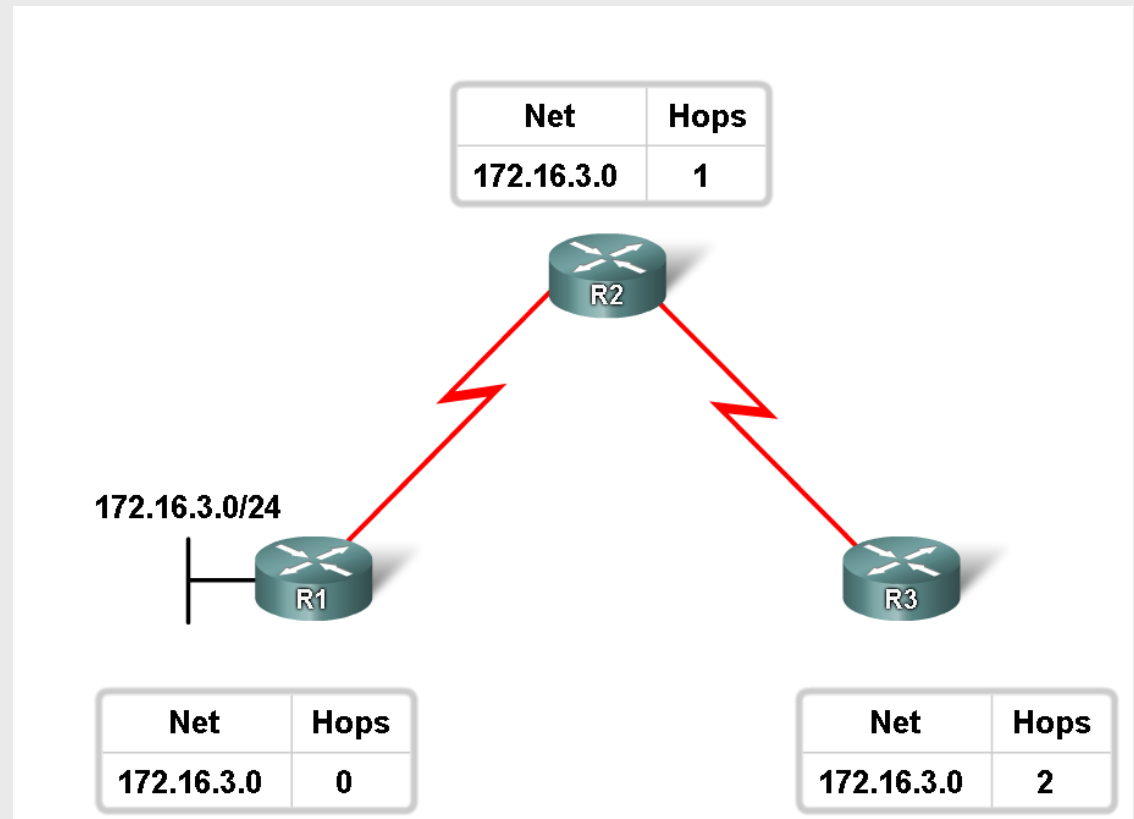
```
C    192.168.2.0/24 is directly connected, Serial0/0/0
```

```
S    192.168.3.0/24 [1/0] via 192.168.2.2
```

```
R    192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:20, Serial0/0/0
```

Routing metrics

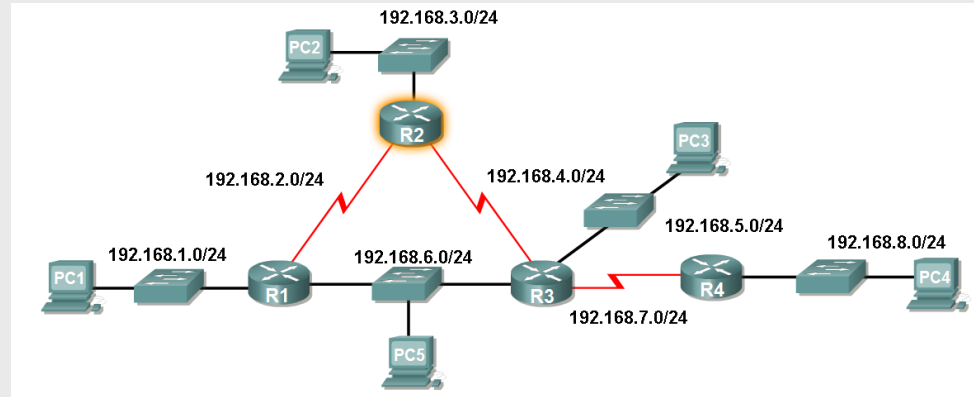
- A quantitative cost of a path to a given destination, needed to compare paths for determining the best one
 - Hop count
 - Bandwidth
 - Cost
 - Delay
 - Load
 - Reliability



Metric field in the routing table

- **Metric** used for each routing protocol

- RIP - hop count
- IGRP & EIGRP - Bandwidth (used by default), Delay (used by default), Load, Reliability
- IS-IS & OSPF – Cost, Bandwidth (Cisco's implementation)



```
R2#show ip route
<output omitted>

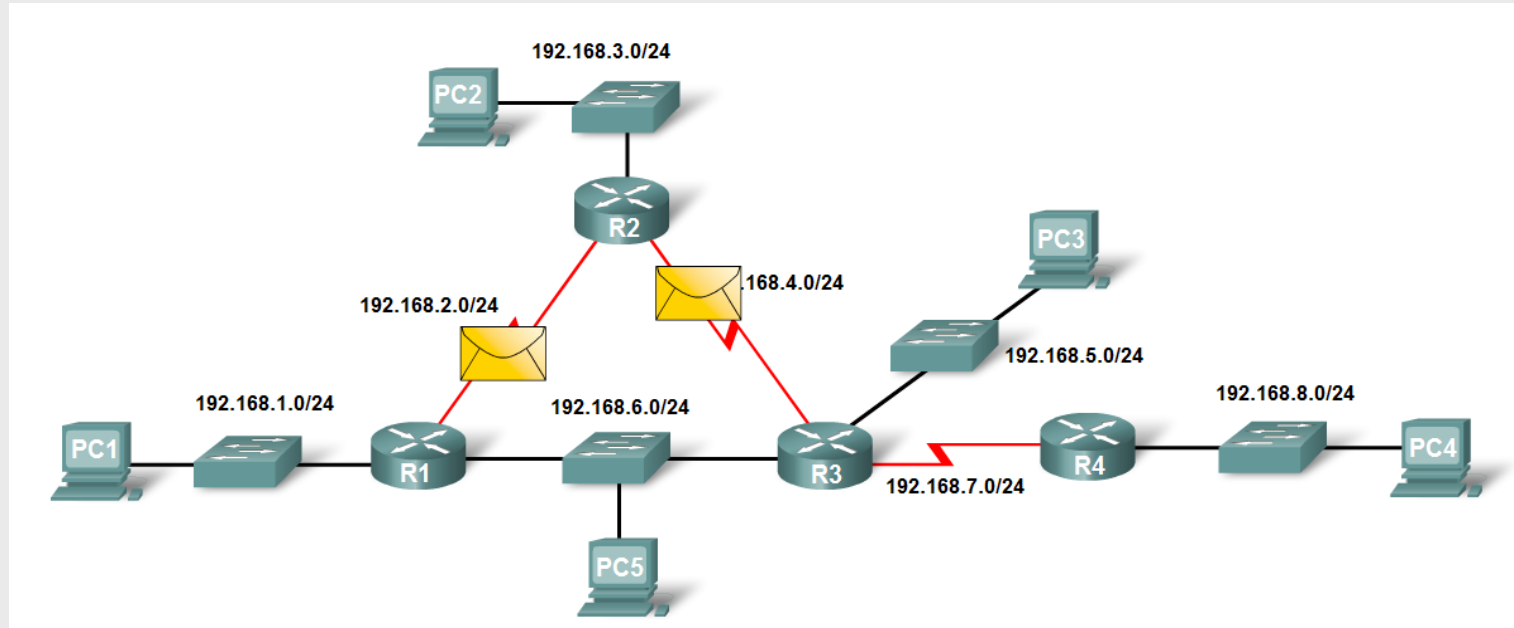
Gateway of last resort is not set

R    192.168.1.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0
C    192.168.2.0/24 is directly connected, Serial0/0
C    192.168.3.0/24 is directly connected, FastEthernet0/0
C    192.168.4.0/24 is directly connected, Serial0/1
R    192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R    192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0
                                     [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R    192.168.7.0/24 [120/1] via 192.168.4.1, 00:00:26, Serial0/1
R    192.168.8.0/24 [120/2] via 192.168.4.1, 00:00:26, Serial0/1
```

It is 2 hops from R2 to 192.168.8.0/24

Load balancing

- Packets are forwarded using all equal-cost paths

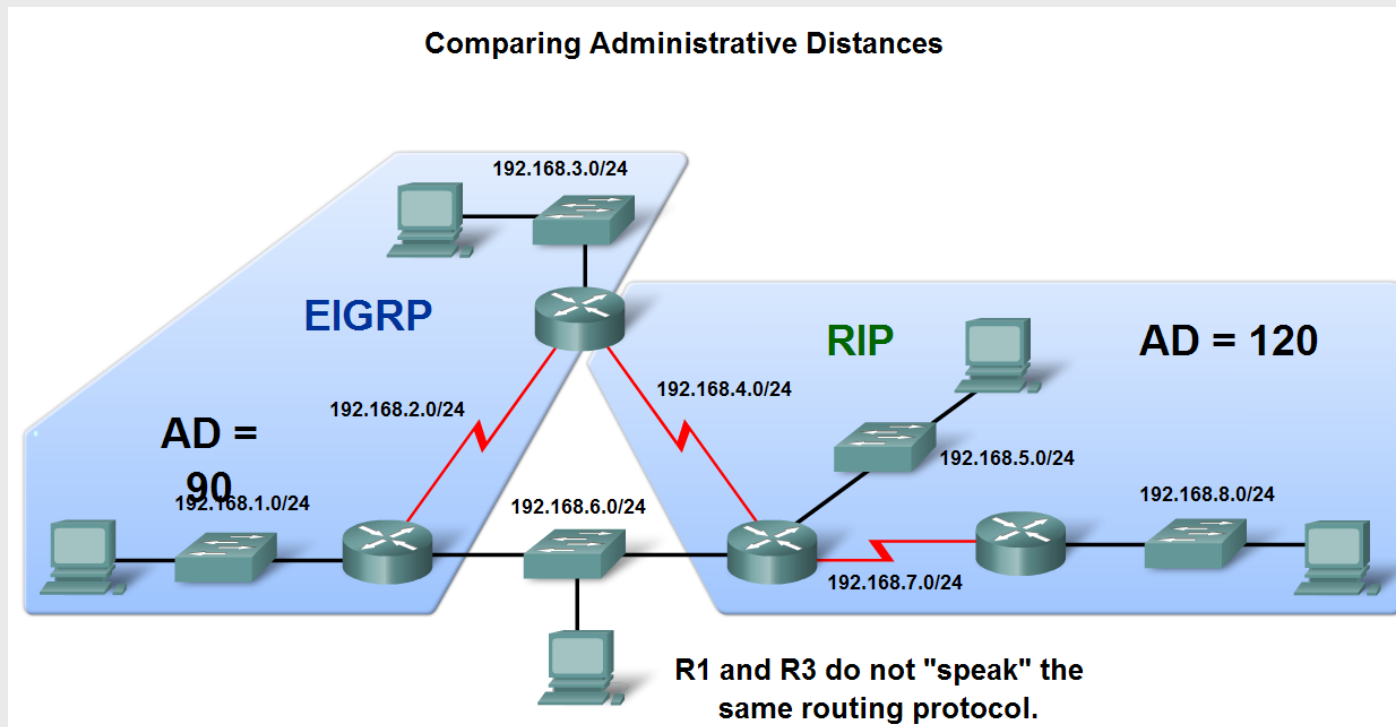


```
R2#show ip route
<output omitted>

R    192.168.6.0/24 [120/1] via 192.168.2.1, 00:00:24, Serial0/0/0
                        [120/1] via 192.168.4.1, 00:00:26, Serial0/0/1
```

Administrative distance

- Administrative Distance (AD) defines the preference of a routing source
 - Integer value from 0 to 255
 - The lower the value the more preferred the route source (the more trustworthy the route)



AD in the routing table

- First number in the brackets in the routing table

```
R2#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
D    192.168.1.0/24 [90/2172416] via 192.168.2.1, 00:00:24, Serial0/0/0  
C    192.168.2.0/24 is directly connected, Serial0/0/0  
C    192.168.3.0/24 is directly connected, FastEthernet0/0  
C    192.168.4.0/24 is directly connected, Serial0/0/1  
R    192.168.5.0/24 [120/1] via 192.168.4.1, 00:00:08, Serial0/0/1  
D    192.168.6.0/24 [90/2172416] via 192.168.2.1, 00:00:24, Serial0/0/0  
R    192.168.7.0/24 [120/1] via 192.168.4.1, 00:00:08, Serial0/0/1  
R    192.168.8.0/24 [120/2] via 192.168.4.1, 00:00:08, Serial0/0/1
```

Default AD

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

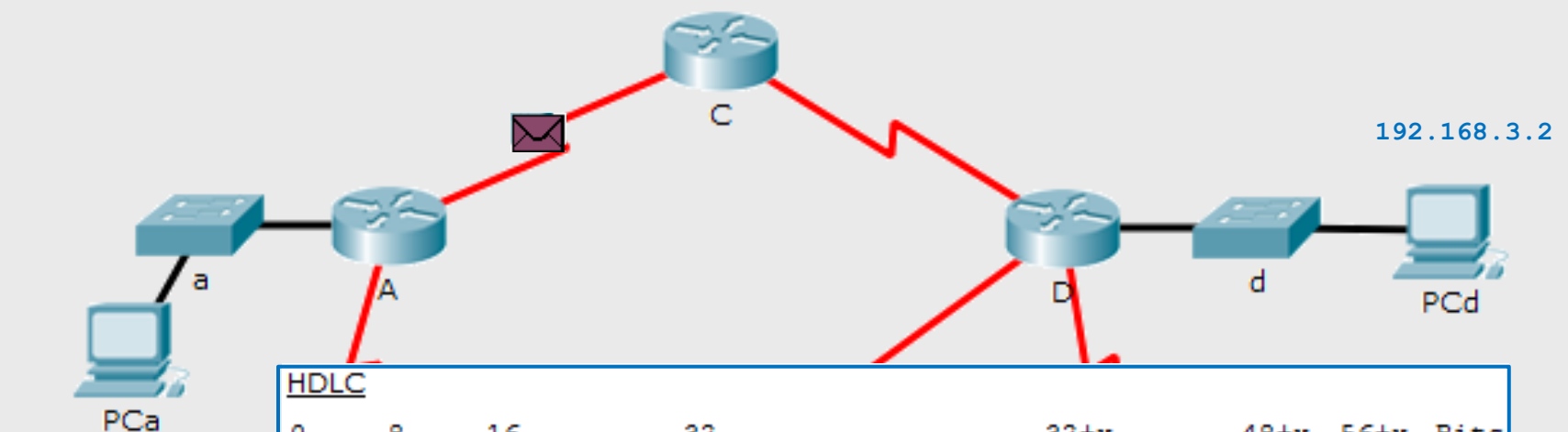


Path determination and switching

- **Path determination** – pick the best path to a destination
 - Directly connected network
 - Remote network
 - No route determined
 - Destination unreachable

- **Switching** – switch a packet from an incoming interface to an outgoing interface
 - The Source and Destination IP addresses **never** change
 - The Source & Destination MAC addresses **change** as packet is forwarded from one router to the next.
 - TTL field decrements by one until a value of zero is reached, at which point the router discards the packet
 - this prevents packets from endlessly traversing the network

Switching



HDLC					
0	8	16	32	32+x	48+x 56+x Bits
FLG: 0111 1110	ADR: 0x8f	CONTROL: 0x0	DATA: (VARIABLE LENGTH)	FCS: 0x0	FLG: 0111 1110

IP

0	4	8	16	19	31	Bits
4	IHL	DSCP: 0x0	TL: 128			
ID: 0x5			0x0	0x0		
TTL: 127		PRO: 0x1		CHKSUM		
SRC IP: 192.168.0.2						
DST IP: 192.168.3.2						
OPT: 0x0					0x0	
DATA (VARIABLE LENGTH)						

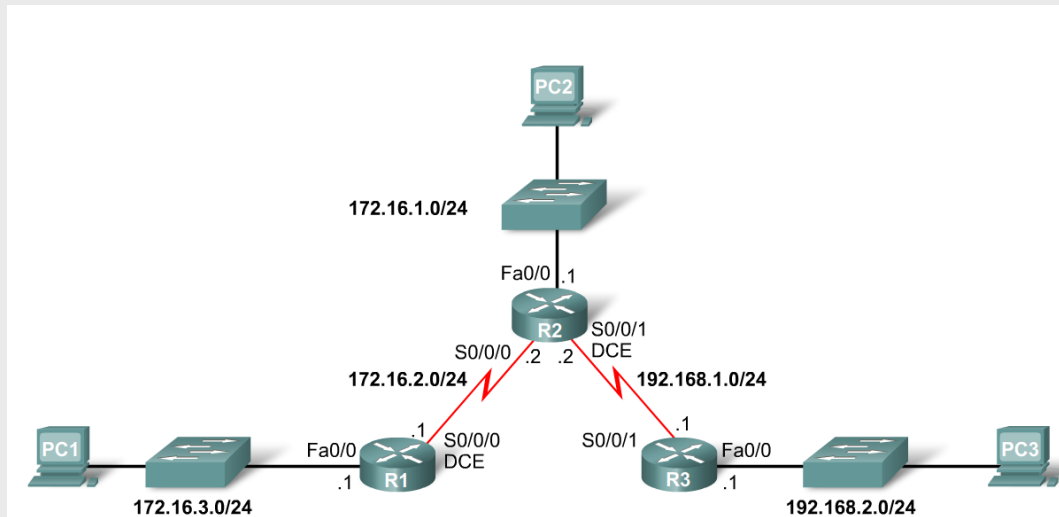
Static route configuration

■ Command **ip route**

```
Router(config)# ip route network-address subnet-mask  
{ip-address | exit-interface } [Distance]
```

Parameter	Description
network-address	Destination network address of the remote network to be added to the routing table.
subnet-mask	Subnet mask of the remote network to be added to the routing table. The subnet mask can be modified to summarize a group of networks.
ip-address	Commonly referred to as the next-hop router's IP address.
exit-interface	Outgoing interface that is used to forward packets to the destination network.

Static route configuration



```
R1#debug ip routing
(**output omitted**)

R1#conf t
R1(config)#ip route 172.16.1.0 255.255.255.0 172.16.2.2

00:20:15: RT: add 172.16.1.0/24 via 172.16.2.2, static metric [1/0]

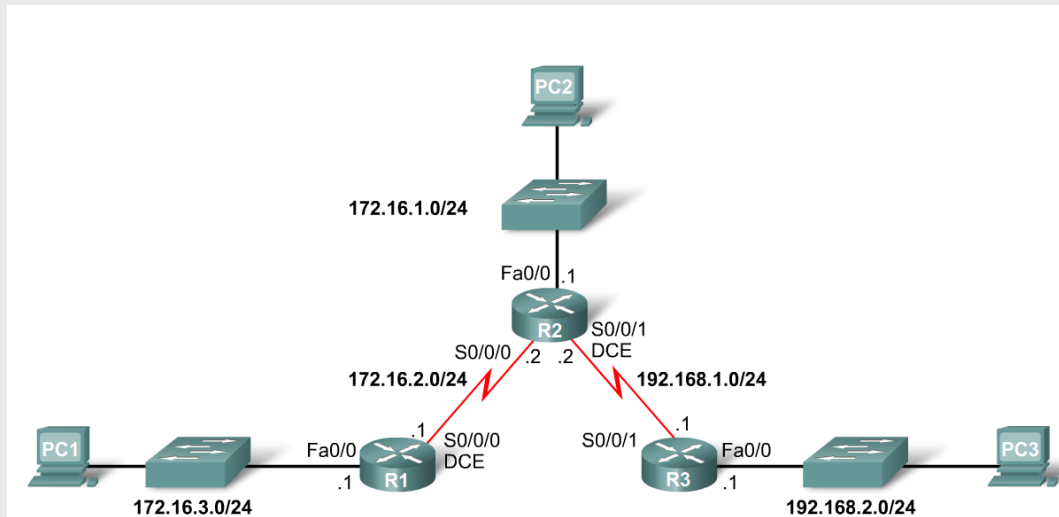
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
R1#
```

Static route configuration

- Configuring routes to two or more remote networks



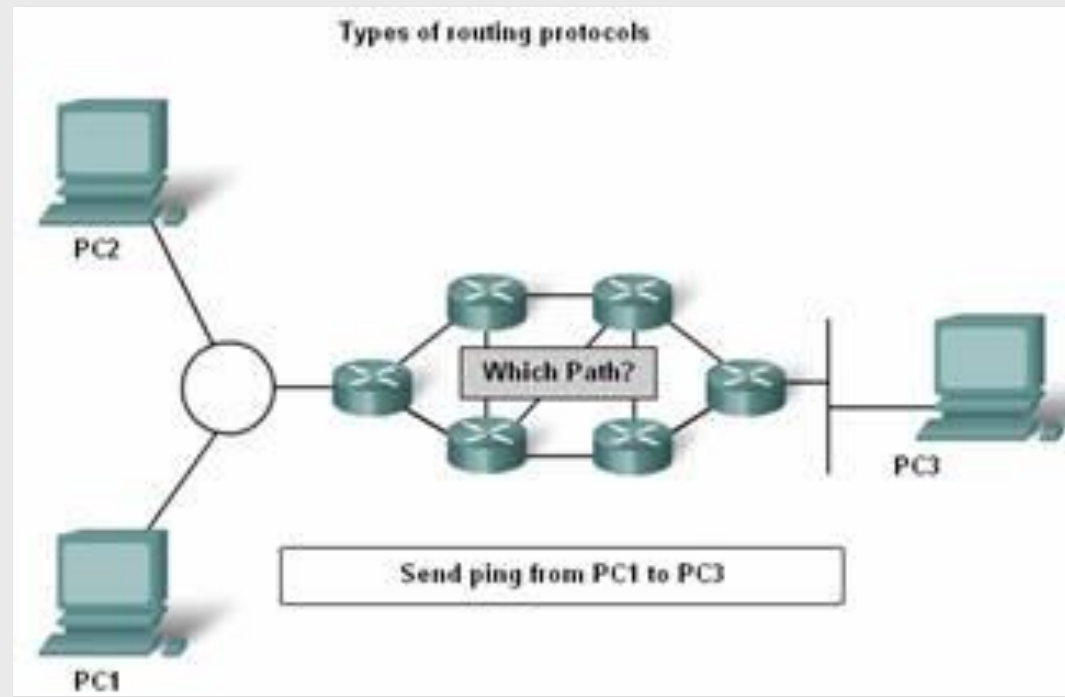
```
R1(config)#ip route 192.168.1.0 255.255.255.0 172.16.2.2
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1(config)#end
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

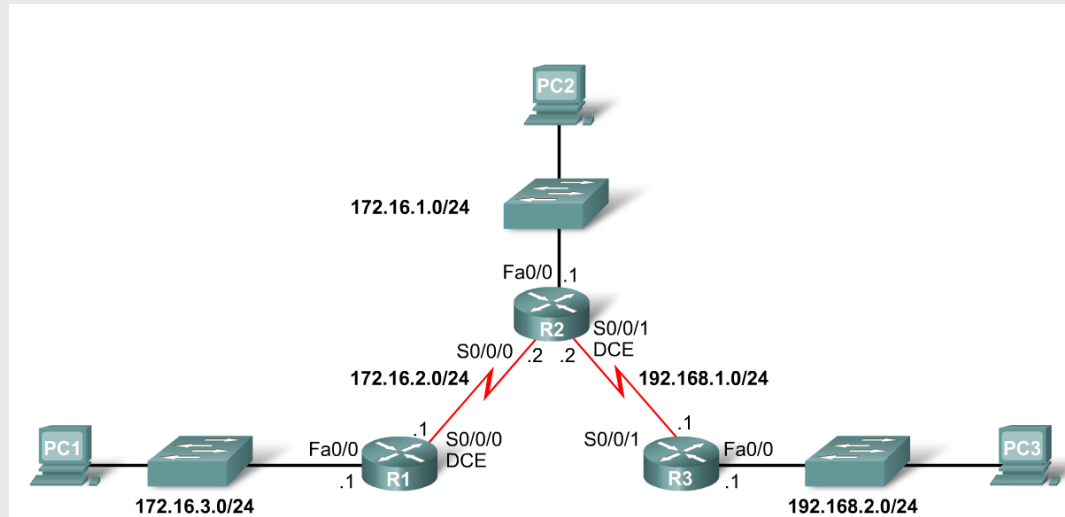
172.16.0.0/24 is subnetted, 3 subnets
S    172.16.1.0 [1/0] via 172.16.2.2
C    172.16.2.0 is directly connected, Serial0/0/0
C    172.16.3.0 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 172.16.2.2
S    192.168.2.0/24 [1/0] via 172.16.2.2
```

Routing table principles

- Every router makes its decisions **alone**, based on the information it has in its routing table.
- Different routing tables may contain **different** information
- A routing table can tell how to get to a destination but **not** how to get back



Recursive route lookup

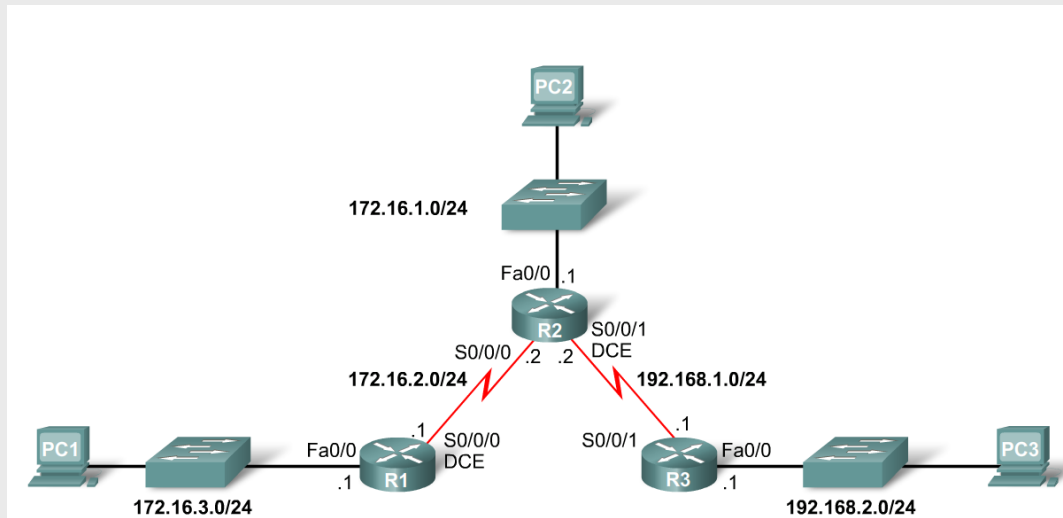


```
R1#show ip route
(**output omitted**)
  172.16.0.0/24 is subnetted, 3 subnets
S    172.16.1.0 [1/0] via 172.16.2.2
C    172.16.2.0 is directly connected, Serial0/0/0
C    172.16.3.0 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 172.16.2.2
S    192.168.2.0/24 [1/0] via 172.16.2.2
```

Step 1: Find a route.

Step 2: Find an exit interface.

Exit interface down



```
R1#debug ip routing
IP routing debugging is on
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s0/0/0
R1(config-if)#shutdown
R1(config-if)#end

is_up: 0 state: 6 sub state: 1 line: 0
RT: interface Serial0/0/0 removed from routing table
RT: del 172.16.2.0/24 via 0.0.0.0, connected metric [0/0]
RT: delete subnet route to 172.16.2.0/24
RT: del 192.168.1.0 via 172.16.2.2, static metric [1/0]
RT: delete network route to 192.168.1.0
RT: del 172.16.1.0/24 via 172.16.2.2, static metric [1/0]
RT: delete subnet route to 172.16.1.0/24

R1#show ip route
***output omitted***
Gateway of last resort is not set
  172.16.0.0/24 is subnetted, 1 subnets
C    172.16.3.0 is directly connected, FastEthernet0/0
```

Static route configuration

- Configure a static route with an exit interface
 - Lookup is more efficient

```
R1(config)#no ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1(config)#ip route 192.168.2.0 255.255.255.0 serial 0/0/0
R1(config)#end
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

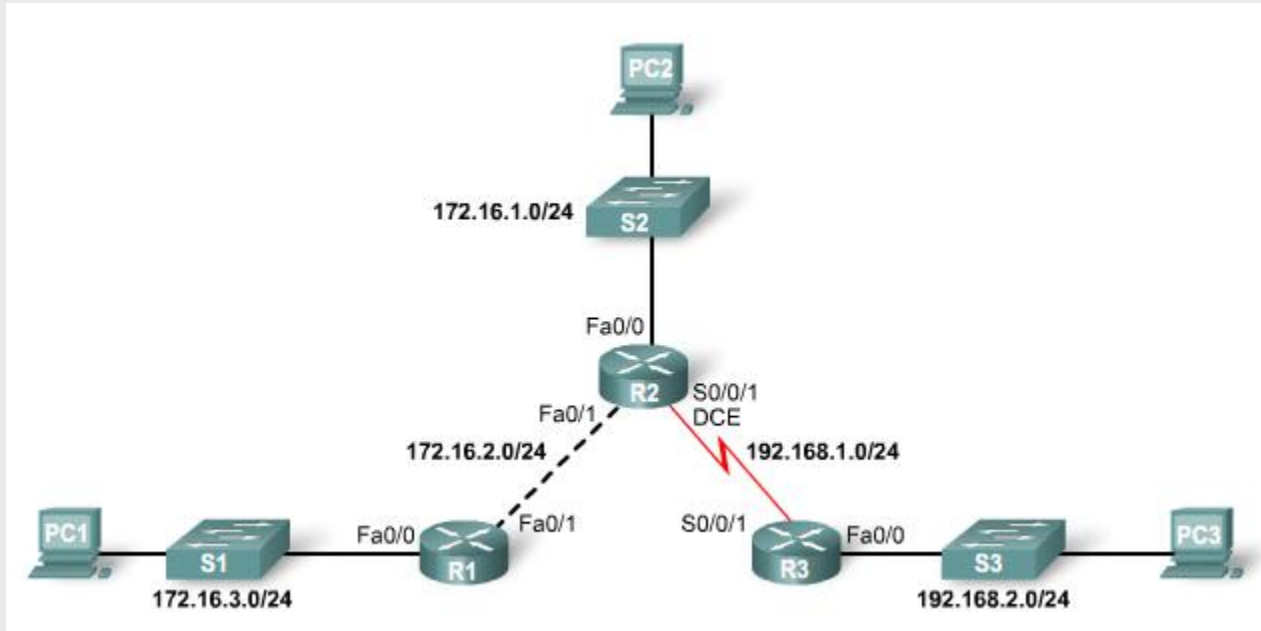
    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 is directly connected, Serial0/0/0
```

Exit interface now specified in the static route. No need for a recursive lookup.

Static route configuration

- Configure a static route with an exit interface
 - Issue with Ethernet exit interfaces

```
R1(config)#ip route 192.168.2.0 255.255.255.0 fastethernet 0/1
```



```
R1(config)#ip route 192.168.2.0 255.255.255.0 fastethernet 0/1 172.16.2.2
```

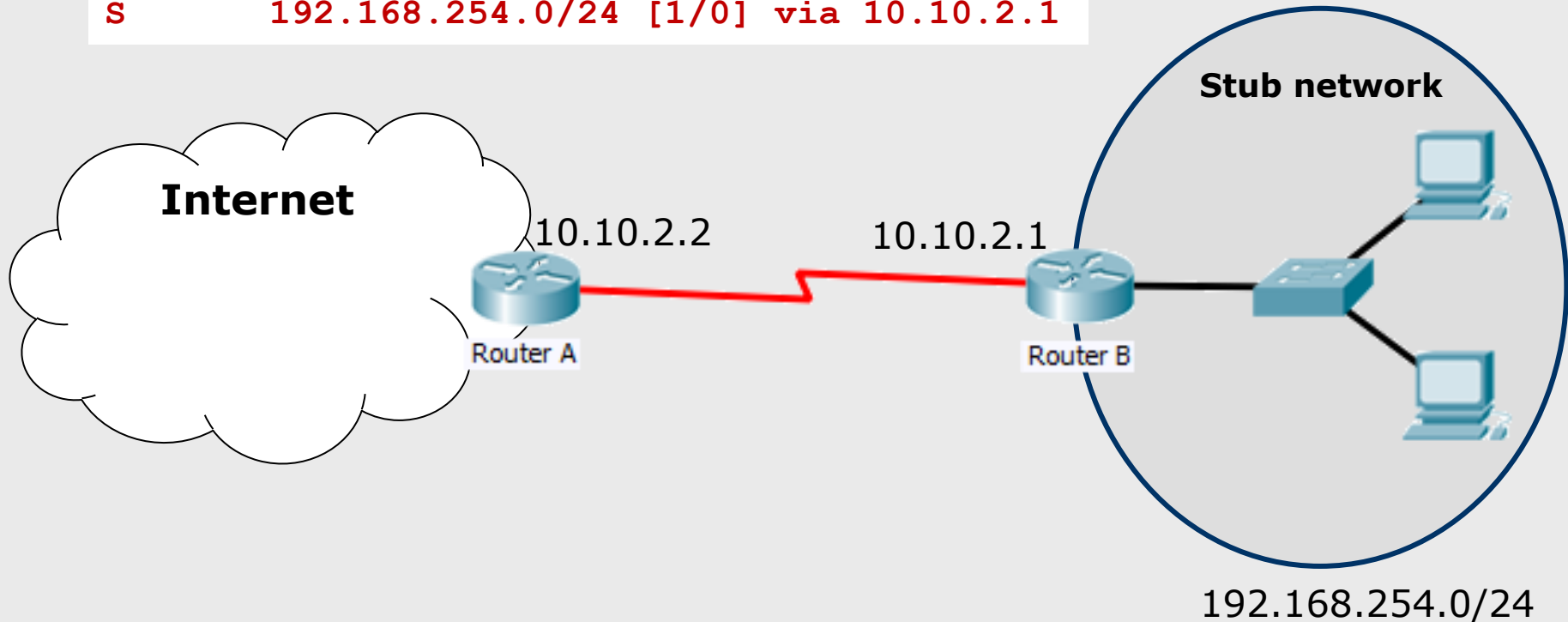
Static routing – when to use

- Dynamic routing is not preferred for security reasons
 - Routing protocol can be a means to compromise network operation
 - Static routing enforces routes
 - To pass through firewalls
 - To pass through safe portions of the network
- Routing policies are in place
 - Service level agreements
 - Load balancing

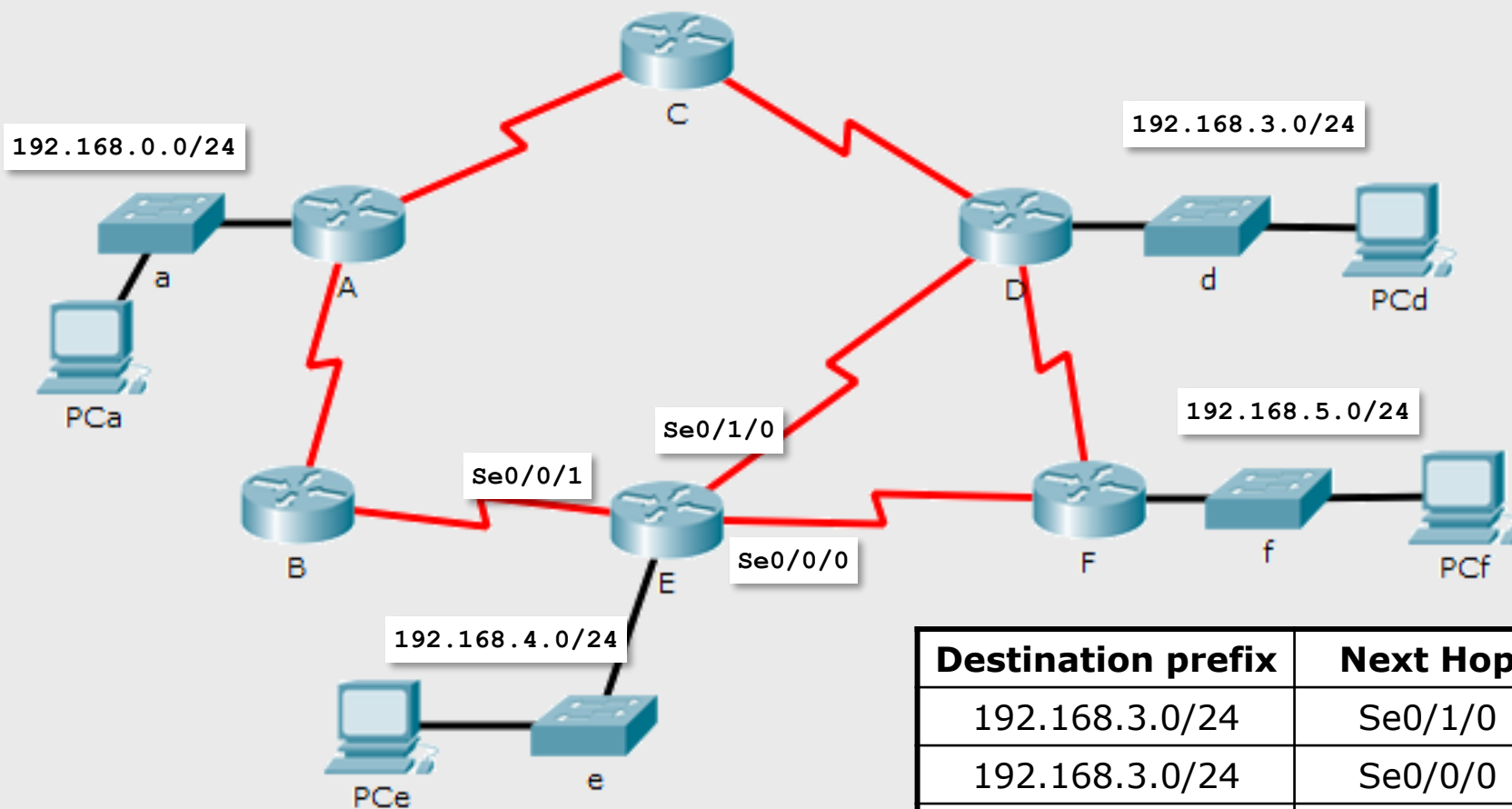
Static routing – example of use

Static route in Router A

S **192.168.254.0/24 [1/0] via 10.10.2.1**



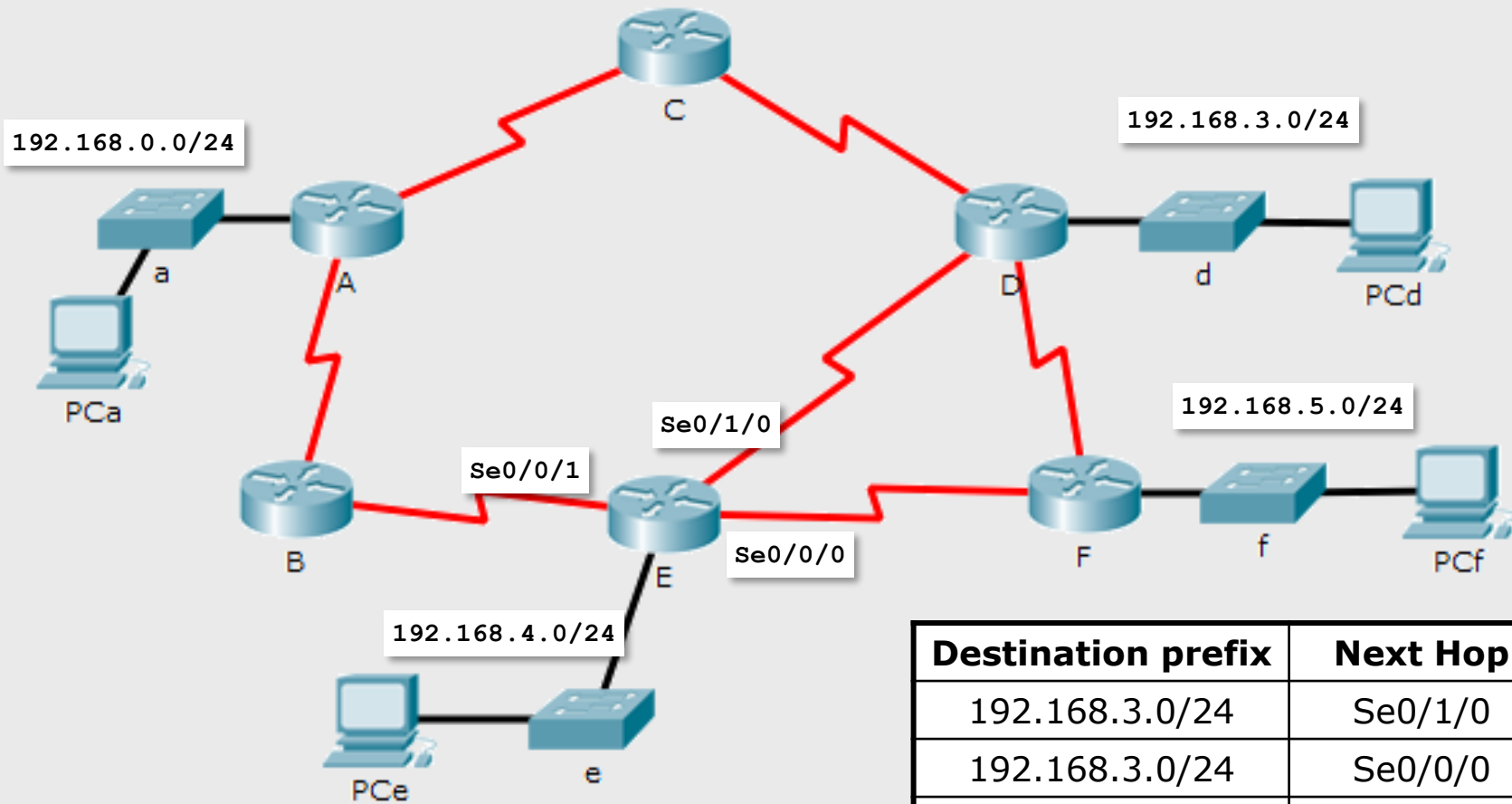
Static routing – design phase



Routes to destinations must be carefully planned on each router **to avoid routing loops**

Destination prefix	Next Hop	Cost
192.168.3.0/24	Se0/1/0	30
192.168.3.0/24	Se0/0/0	40
192.168.0.0/24	Se0/0/1	30
192.168.0.0/24	Se0/1/0	40
192.168.5.0/24	Se0/0/0	30

Static routing – configuration phase

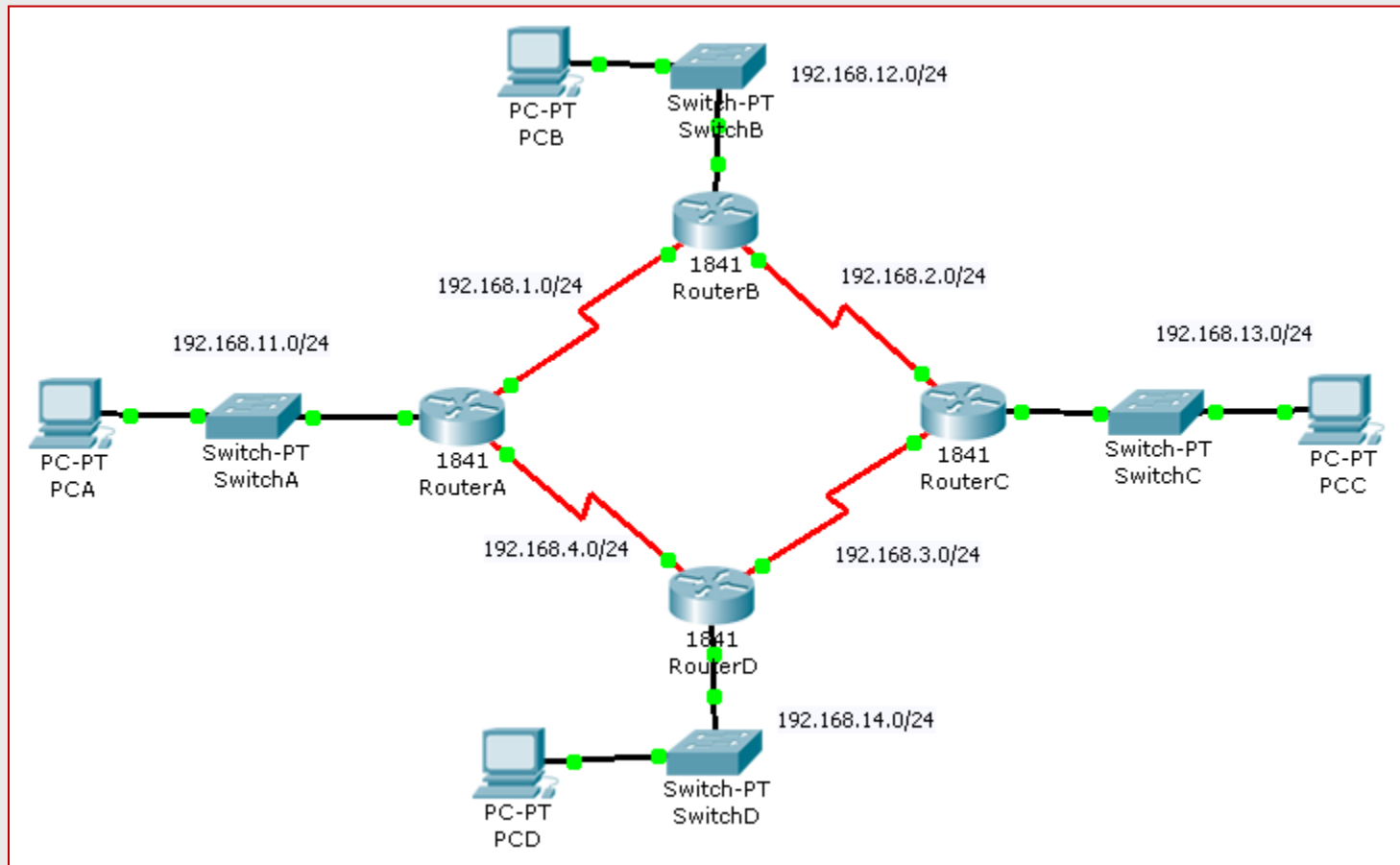


Routing table entries are manually configured by the network manager (already covered)

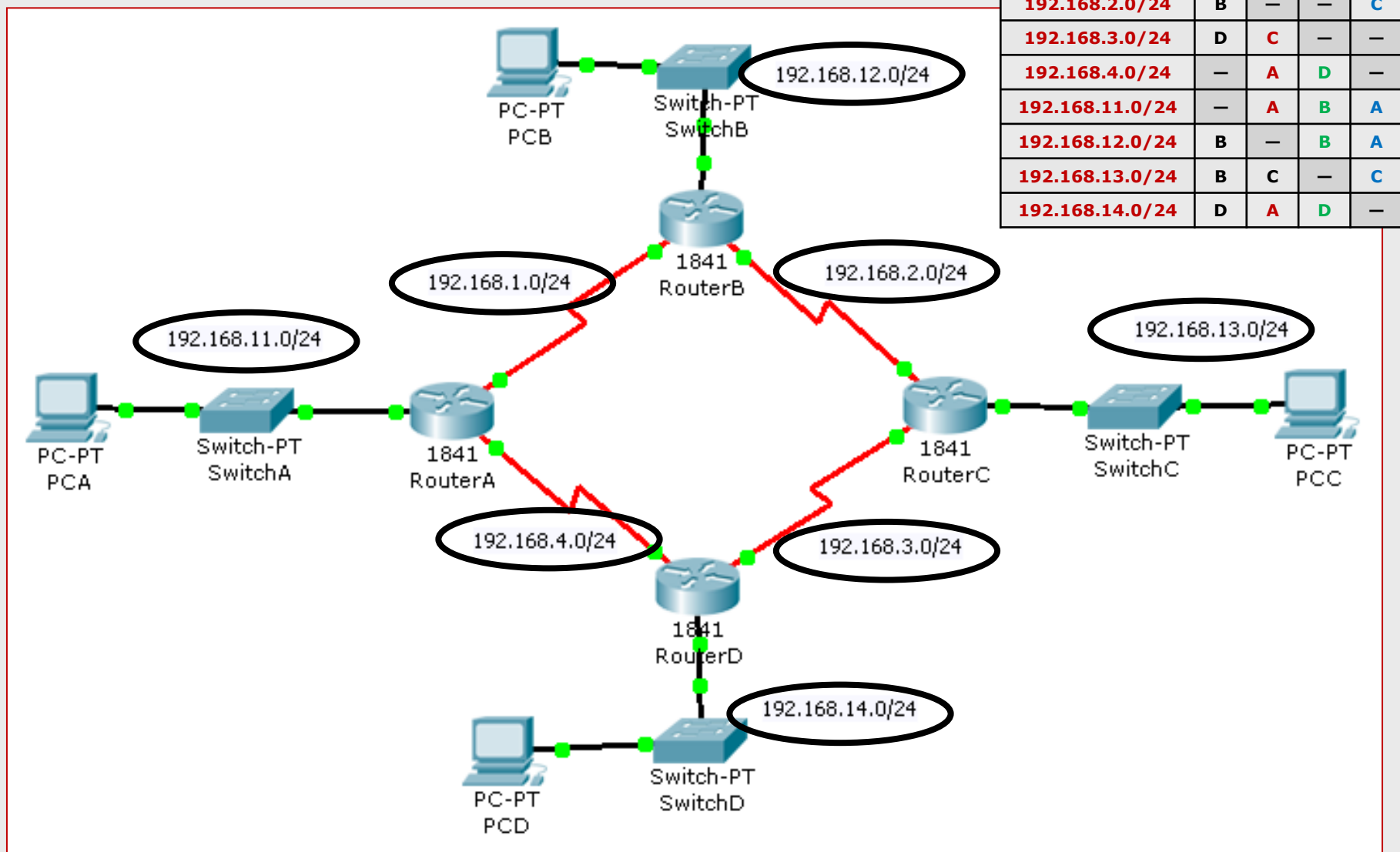
Destination prefix	Next Hop	Cost
192.168.3.0/24	Se0/1/0	30
192.168.3.0/24	Se0/0/0	40
192.168.0.0/24	Se0/0/1	30
192.168.0.0/24	Se0/1/0	40
192.168.5.0/24	Se0/0/0	30

Static routing – design phase

- Use of a “design table” to keep track of planned routes and identify loops



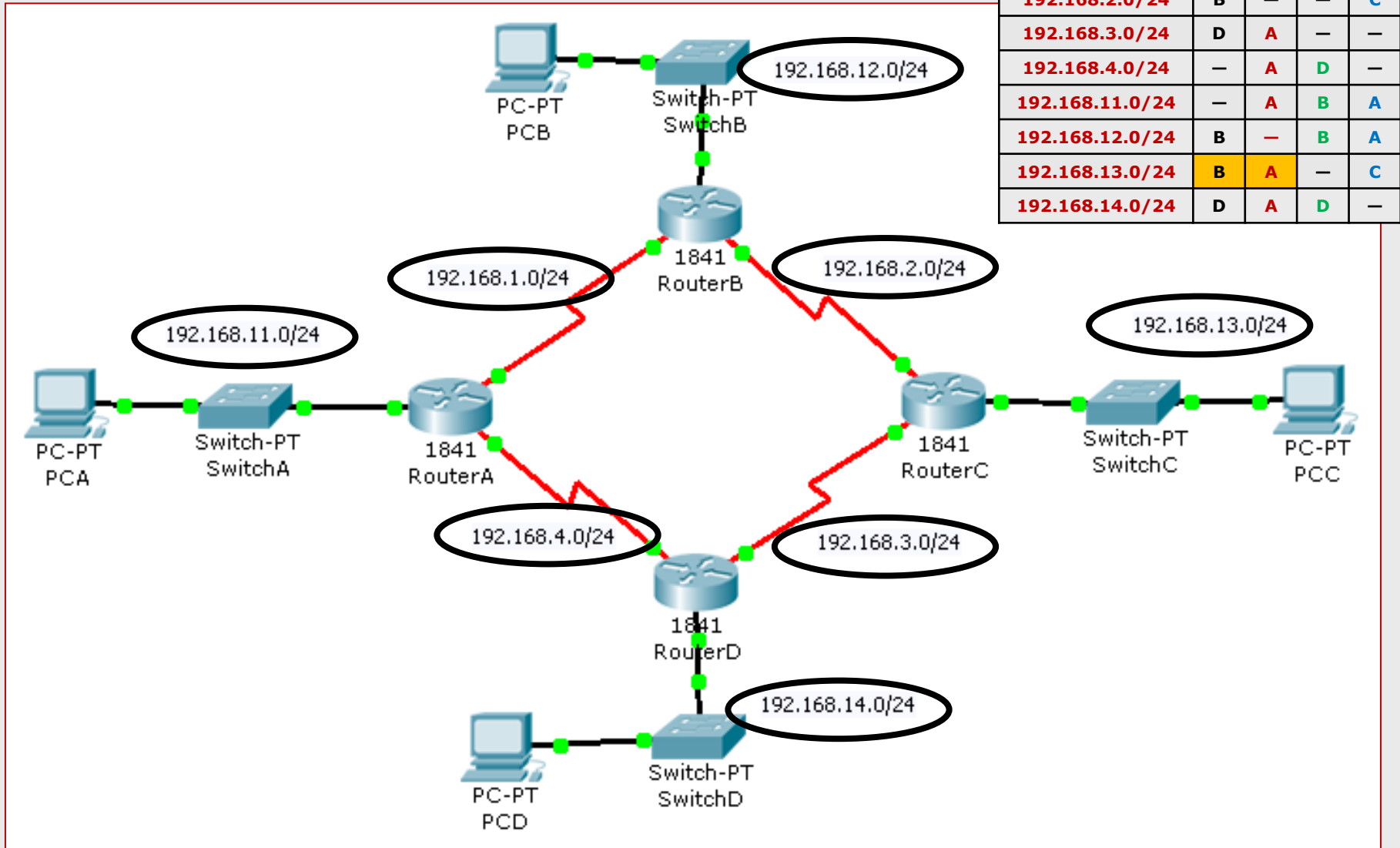
Static routing – design phase



Static routing – design phase

■ RouterB

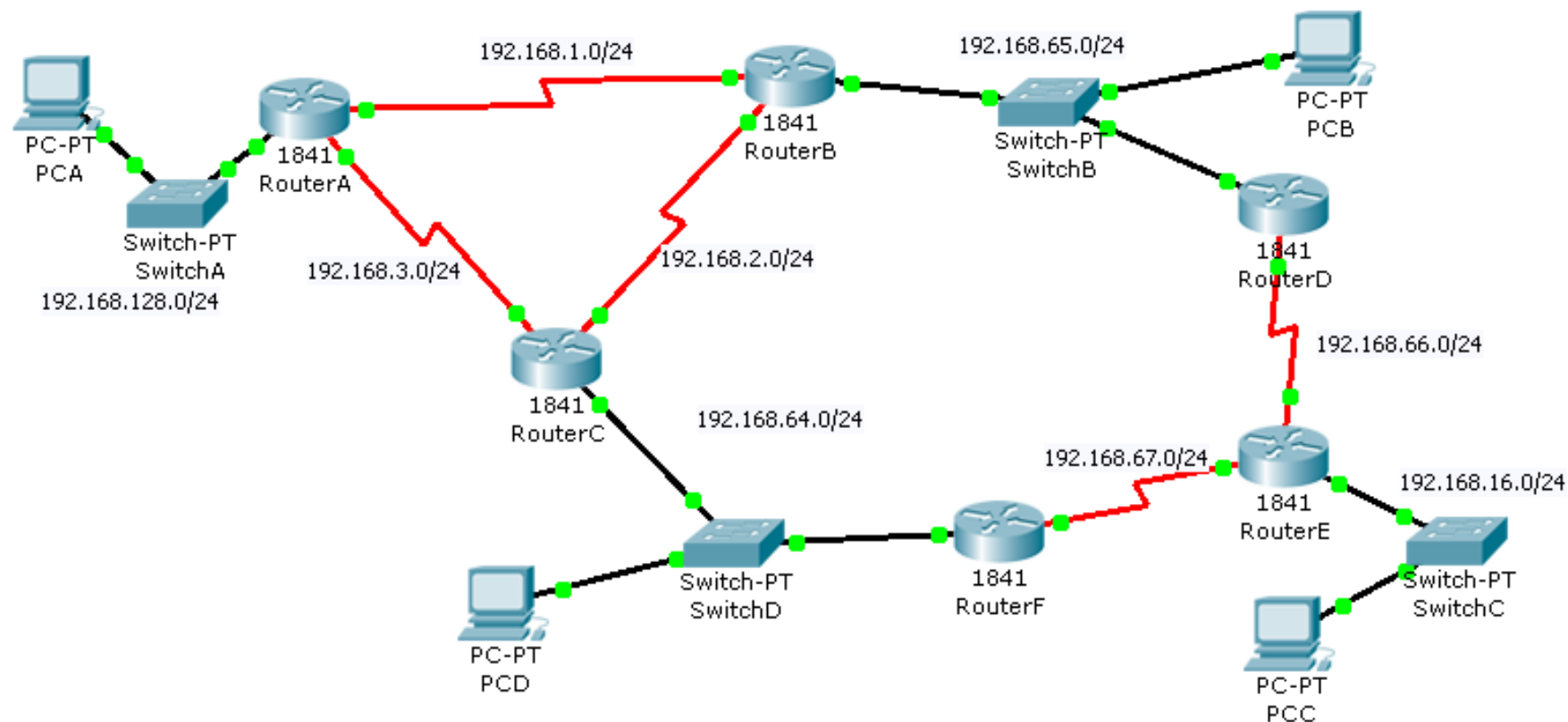
Routing Loop



Static routing – design phase

- How to avoid loops?
 - Loops are **guaranteed** to be avoided if the shortest-path tree to each destination network is used for each router
 - For directed graphs with non-negative weights, the shortest-path tree is computed by the **Dijkstra's algorithm**
- What if additional constraints are specified?
 - e.g., do not use a specific link for primary paths, ...
 - A constrained shortest-path tree is defined
 - out of scope of this class

Static routing – design phase



Static routing – design phase

- Use of a “design table” to keep track of planned routes and identify loops

Network	RA	RB	RC	RD	RE	RF
192.168.128.0/24	—					
192.168.1.0/24	—	—				
192.168.2.0/24		—	—			
192.168.3.0/24	—		—			
192.168.16.0/24					—	
192.168.64.0/24			—			—
192.168.65.0/24		—		—		
192.168.66.0/24				—	—	
192.168.67.0/24					—	—

Static routing – design phase

- Use of a “design table” to keep track of planned routes and identify loops

Network	RA	RB	RC	RD	RE	RF
192.168.128.0/24	—	RA	RA	RB	RF	RC
192.168.1.0/24	—	—	RB	RB	RD	RC
192.168.2.0/24	RB	—	—	RB	RF	RC
192.168.3.0/24	—	RA	—	RB	RF	RC
192.168.16.0/24	RC	RD	RF	RE	—	RE
192.168.64.0/24	RC	RC	—	RB	RF	—
192.168.65.0/24	RB	—	RB	—	RD	RC
192.168.66.0/24	RB	RD	RB	—	—	RC
192.168.67.0/24	RC	RC	RF	RB	—	—